Charlie Zhang

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EDUCATION

Columbia University
Master of Arts in Statistics (Advanced Machine Learning Track) — *Expected December 2026*
Coursework: Machine Learning, Deep Learning, Statistical Learning, Algorithms for DS, NLP, Generative AI & LLMs

University of Minnesota, Twin Cities
Bachelor of Science in Statistical Science & Bachelor of Science in Economics | GPA: 3.6 / 4.0 — *August 2025*
Coursework: Stat. Machine Learning I/II, Grad-Level Stat. Theory I/II, Stat. Computing (Python/R), Adv. Math Econ

WORK EXPERIENCE

Columbia University **| *Graduate Research Assistant*** — *Jan 2026 - Present | 12h/week*

- Collaborate with the Social Intervention Group (SIG) and AI4SGS to benchmark LLM-based qualitative coding tools against human-coded gold standards using quantitative agreement metrics (e.g., κ, F1); develop reproducible evaluation pipelines in Python.
- Conduct data cleaning, statistical analysis, and literature reviews for community-engaged intervention research; deliver research materials and internal reports for interdisciplinary teams.

Minsheng Securities **| *Quantitative Analysis Internship*** — *June 2024 - July 2024*

- Engineered automated data pipelines in Python to collect and clean large-scale financial datasets (~100,000+ records), ensuring high data integrity for predictive modeling.
- Built regression and time-series forecasting models to predict short-term market returns, achieving 10–15% improvement in predictive accuracy compared to baseline.

Minsheng Securities **| *Data Analysis Internship*** — *June 2023 - July 2023*

- Synthesized technology sector dynamics and financial indicators to perform quantitative risk assessments, delivering data-driven insights that informed strategic investment decisions.
- Contributed to a data-driven allocation strategy that outperformed the market benchmark by 5%, demonstrating the ability to translate raw data into business value.

Institute for Research in Statistics and its Applications (IRSA)**| *Statistical Consulting Intern*** — *May 2024 - September 2024*

- Performed multivariate analysis (MANOVA) in R on 100+ Spanish speech samples, building statistical models to evaluate the effects of group membership, gender, and vowel stress on acoustic outcomes (F1, F2, duration).
- Generated actionable insights from complex linguistic data, uncovering significant variation patterns in Spanish vowel production and translating results into recommendations on bilingual language use and community integration.

RESEARCH & PROJECT

RAG with PDF Review | *Applied Machine Learning Research* — *February 2025 - May 2025*

- Developed a retrieval-augmented generation (RAG) PDF query system that parses and chunks PDFs, embeds with OpenAI, and indexes in FAISS for efficient information retrieval.
- Engineered automatic in-document highlighting and deployed a Streamlit web application, packaging the system into a reproducible Python library (open-source on GitHub) to support research reproducibility and broader community adoption.
- Enhanced document query accuracy with re-ranking algorithms, enabling users to extract top-k, page-anchored results and improving relevance scores for downstream tasks such as academic literature review, compliance checks, and research.

Medicaid Expansion Impact Analysis | *Data Analysis Project* — *February 2025 - May 2025*

- Built a state-level panel dataset (1997–2007, ~100k records) by merging CDC mortality, Census income/poverty, and BLS unemployment data.
- Estimated DiD/panel models with robustness checks; quantified a 2-3% decline in mortality affecting 200k+ residents.

Decagon Polypharmacy Graph Learning | *Machine Learning Research Project* — *September 2025 - December 2025*

- Benchmarked baseline, RESCAL, and R-GCN on Decagon (963 side effects; hierarchical disease categories).
- Implemented an R-GCN model in PyTorch Geometric with ChemBERTa and GROVER molecular embeddings; evaluated models using AUROC/AUPR, AP@50, and F1, emphasizing robustness and failure modes under severe class imbalance.

GRANTS, HONORS & SCHOLARSHIPS

Lynn Lin Undergraduate Internship, *University of Minnesota, Twin Cities* — *Summer 2024*

Dean's List, *College of Liberal Arts, University of Minnesota, Twin Cities* — *2023 - 2025*

SKILLS

- **Programming & Data Analysis:** Python (NumPy, Pandas, scikit-learn), R Studio, SQL, Excel (advanced), Git
- **Applied Machine Learning:** Predictive Modeling, Similarity-Based Methods, Feature Engineering; PyTorch, TensorFlow
- **Statistics & Econometrics:** Regression, Time Series, Causal Inference, Experimental Design (A/B Testing)